

RAWLINS, WY, USA

WMO: 726690

Lat: 41.806N

Lon: 107.200W

Elev: 6775

StdP: 11.44

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24057

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.8	-0.6	-13.4	3.4	-6.0	-7.5	4.7	1.6	38.8	29.0	35.7	29.2	5.2	60	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.0	89.6	55.8	87.2	55.2	84.6	54.5	59.3	78.7	58.1	77.3	56.9	76.3	13.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	80.7	62.2	52.1	74.4	61.5	49.7	68.2	62.0	28.9	78.1	28.0	76.9	27.1	76.3	64.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.0	30.8	27.3	DB	-21.4	93.7	6.9	2.1	-26.3	95.2	-30.3	96.4	-34.2	97.6	-39.2	99.1	(4)
(5)			WB	-21.5	61.8	6.2	1.3	-26.0	62.8	-29.6	63.5	-33.1	64.2	-37.7	65.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	22.7	24.0	33.7	40.9	49.8	60.9	69.3	66.1	57.2	43.8	31.8	22.0		
	DBStd	18.56	10.31	10.74	9.97	8.15	8.14	6.68	4.25	4.90	6.98	9.72	10.93	10.52		
	HDD50	4143	848	727	507	284	106	7	0	0	20	229	548	867		
	HDD65	8067	1313	1147	970	722	473	153	14	45	245	656	997	1332		
	CDD50	1820	0	0	2	13	99	334	599	498	236	38	1	0		
	CDD65	267	0	0	0	0	1	29	148	78	11	0	0	0		
	CDH74	5435	0	0	0	1	85	1012	2473	1474	385	5	0	0		
	CDH80	1865	0	0	0	0	9	326	1021	449	60	0	0	0		
(14) Wind	WSAvg	12.3	14.4	14.7	13.8	13.4	11.6	11.1	9.2	9.3	10.0	12.3	13.2	14.2		
(15) Precipitation	PrecAvg	9.7	0.6	0.5	0.6	1.1	1.3	0.9	0.8	0.8	0.9	0.9	0.6	0.6		
	PrecMax	11.8	1.9	1.1	1.9	2.5	2.9	2.0	2.6	2.1	3.6	3.1	2.0	1.3		
	PrecMin	5.9	0.0	0.1	0.2	0.2	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0		
	PrecStd	1.7	0.4	0.2	0.4	0.7	0.8	0.7	0.7	0.6	0.8	0.8	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	51.2	65.6	71.4	80.8	90.9	93.4	89.8	85.3	73.7	61.6	46.9		
		MCWB	34.8	38.2	43.5	46.2	51.8	54.6	57.2	56.0	55.1	48.2	42.8	36.1		
	2%	DB	41.5	45.2	59.6	66.4	76.1	87.2	90.7	87.1	81.4	68.9	56.6	42.9		
		MCWB	33.0	34.9	41.2	43.8	49.8	54.1	56.3	54.9	52.2	47.3	40.6	33.9		
	5%	DB	38.7	41.3	54.2	63.0	72.6	84.0	88.5	84.8	78.8	64.9	52.1	39.3		
		MCWB	31.6	33.0	38.9	42.5	48.7	53.1	55.9	54.4	51.3	45.5	38.6	31.9		
	10%	DB	36.3	38.2	49.4	58.6	68.3	80.5	86.1	82.4	75.3	60.8	47.9	36.5		
		MCWB	30.3	31.4	36.9	41.2	47.3	52.0	55.4	53.9	49.8	43.7	36.6	30.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	39.4	44.3	47.4	54.6	58.2	61.0	60.6	57.4	51.1	44.4	37.2		
		MCDB	42.8	49.4	62.9	66.9	74.3	80.6	79.7	77.3	77.1	67.1	57.9	45.2		
	2%	WB	34.4	35.9	41.8	45.3	51.6	56.1	59.7	58.9	55.2	48.9	41.4	34.8		
		MCDB	39.8	43.3	57.8	62.7	70.8	79.0	79.5	76.6	72.6	64.6	54.5	41.1		
	5%	WB	32.4	33.9	39.5	43.5	50.1	54.8	58.7	57.4	53.4	46.7	39.3	32.8		
		MCDB	37.7	40.3	52.3	60.2	68.1	77.4	78.5	75.3	70.7	62.4	50.9	38.6		
	10%	WB	30.6	32.0	37.2	41.7	48.3	53.3	57.7	56.2	52.0	44.6	37.1	30.7		
		MCDB	35.8	37.9	48.4	57.3	65.5	75.4	77.0	74.7	69.1	59.0	46.8	35.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.3	20.9	24.4	26.8	31.9	32.0	30.6	29.5	24.0	19.4	16.1		
		MCDBR	16.7	18.0	27.3	33.1	34.3	38.1	36.3	34.9	35.3	30.5	25.0	17.9		
		MCWBR	11.2	11.5	13.7	15.7	14.2	14.5	12.7	12.6	14.8	14.9	13.6	11.9		
		MCDBR	15.3	15.5	25.7	29.7	31.1	33.5	30.3	28.0	27.9	27.9	23.8	16.3		
	5% WB	MCWBR	10.8	10.3	13.4	14.8	13.5	13.1	10.2	10.3	11.4	13.9	13.2	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.225	0.233	0.255	0.288	0.303	0.305	0.332	0.331	0.295	0.257	0.237	0.229		
	taud		2.511	2.518	2.527	2.459	2.473	2.475	2.430	2.421	2.521	2.616	2.580	2.519		
	Ebn at Noon		305	318	320	312	308	306	296	294	300	305	298	294		
	Edh at Noon		22	26	29	34	34	34	36	35	29	23	20	20		
(44) All-Sky Solar Radiation	RadAvg		682	964	1374	1735	2035	2408	2309	2013	1659	1138	747	551		
	RadStd		45	60	114	115	149	134	79	62	108	77	69	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page